

EDA ON E-COMMERCE CUSTOMER INTERACTION DATASET

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PROJECT OVERVIEW

The objective of this project is to analyze customer interaction data from marketing campaigns and identify customer engagement patterns using Exploratory Data Analysis (EDA).

Business Goal

- Measure campaign performance
- Analyze customer engagement
- Evaluate open and click behavior
- Generate business insights for marketing optimization

14.94

CTR

39.86

Open Rate

86K

Total Customers

200K

Count of CampaignID

DATASET OVERVIEW

Dataset Information

- Total Rows: 200,000
- Total Columns: 4
- Missing Values: 0
- Duplicate Rows: 0
- Dataset Size: 6.1 MB

```
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CampaignID      200000 non-null  int64
1   CustomerID      200000 non-null  int64
2   Opened          200000 non-null  int64
3   Clicked         200000 non-null  int64
dtypes: int64(4)
memory usage: 6.1 MB
```

DATA CLEANING & QUALITY CHECK

Data Quality Findings

- No missing values found
- No duplicate records
- Binary engagement columns properly formatted
- Dataset is clean and analysis-ready

Importance

High-quality data improves:

- Analytics accuracy
- Dashboard reliability
- ML model performance

CUSTOMER ANALYSIS

Customer Insights

- Total unique customers: 86,438
- Customers interacted multiple times across campaigns
- Indicates repeat engagement behavior

Business Interpretation

This data can support:

- Customer segmentation
- Retention analysis
- Personalized marketing strategies

OPEN RATE ANALYSIS

Open Rate Results

- Opened Campaigns: 79,724
- Not Opened: 120,276
- Open Rate: 39.86%

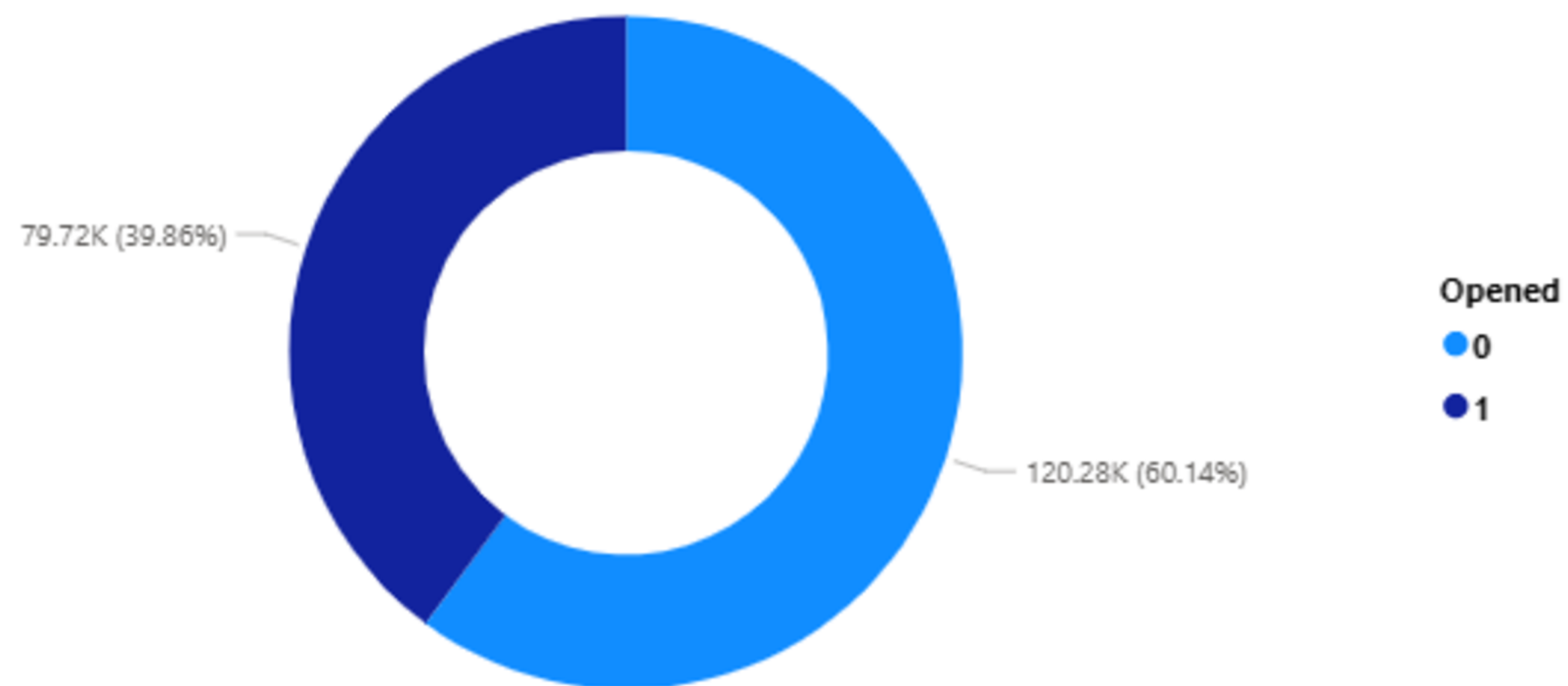
Insight

The open rate is higher than industry average email campaign performance.

Business Meaning

- Effective targeting
- Strong subject lines
- Engaged customer base

Clicked vs Not Clicked



CLICK THROUGH RATE (CTR) ANALYSIS

CTR Results

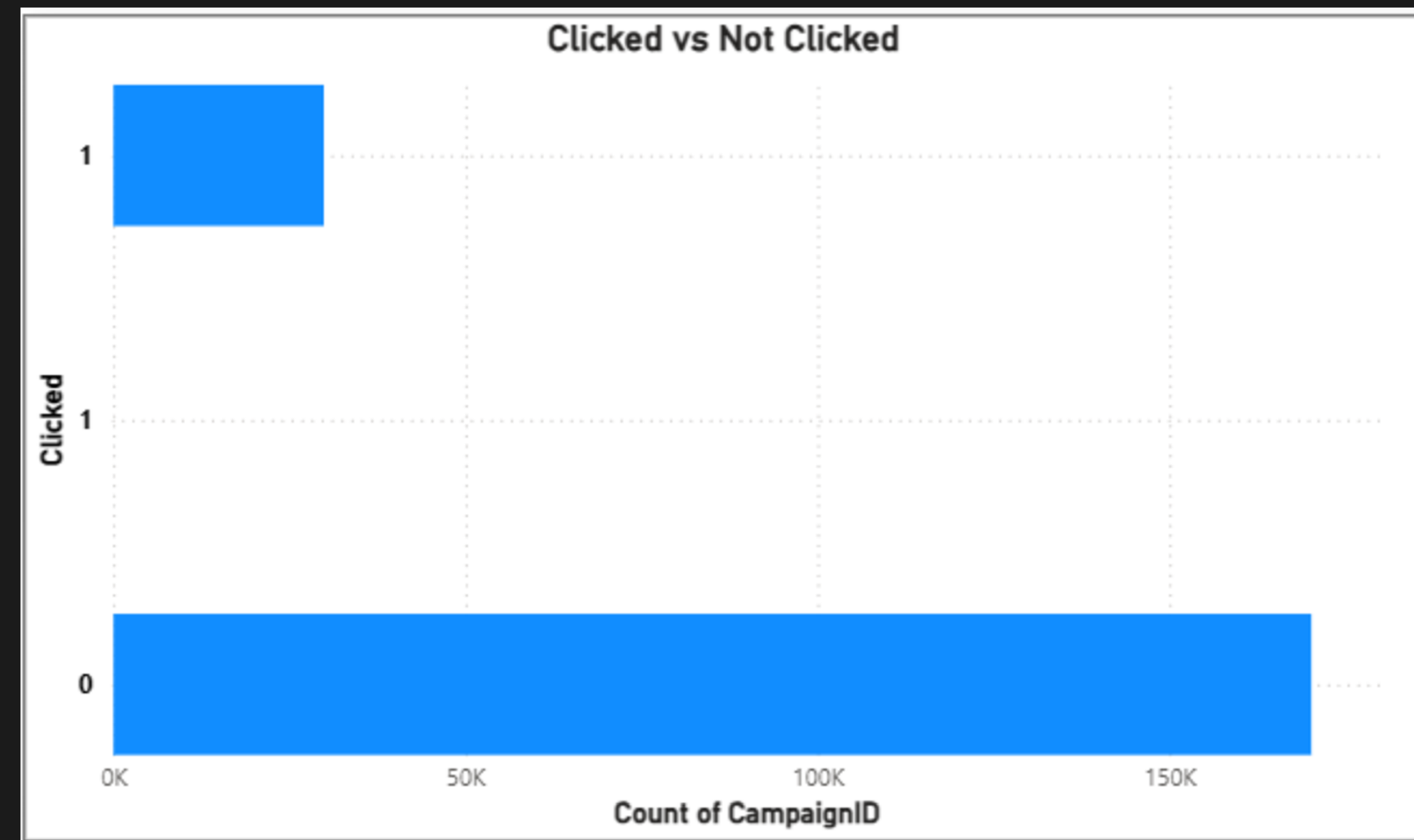
- Clicked Campaigns: 29,876
- Not Clicked: 170,124
- CTR: 14.94%

Insight

Customers showed strong interest in campaign offers and products.

Business Meaning

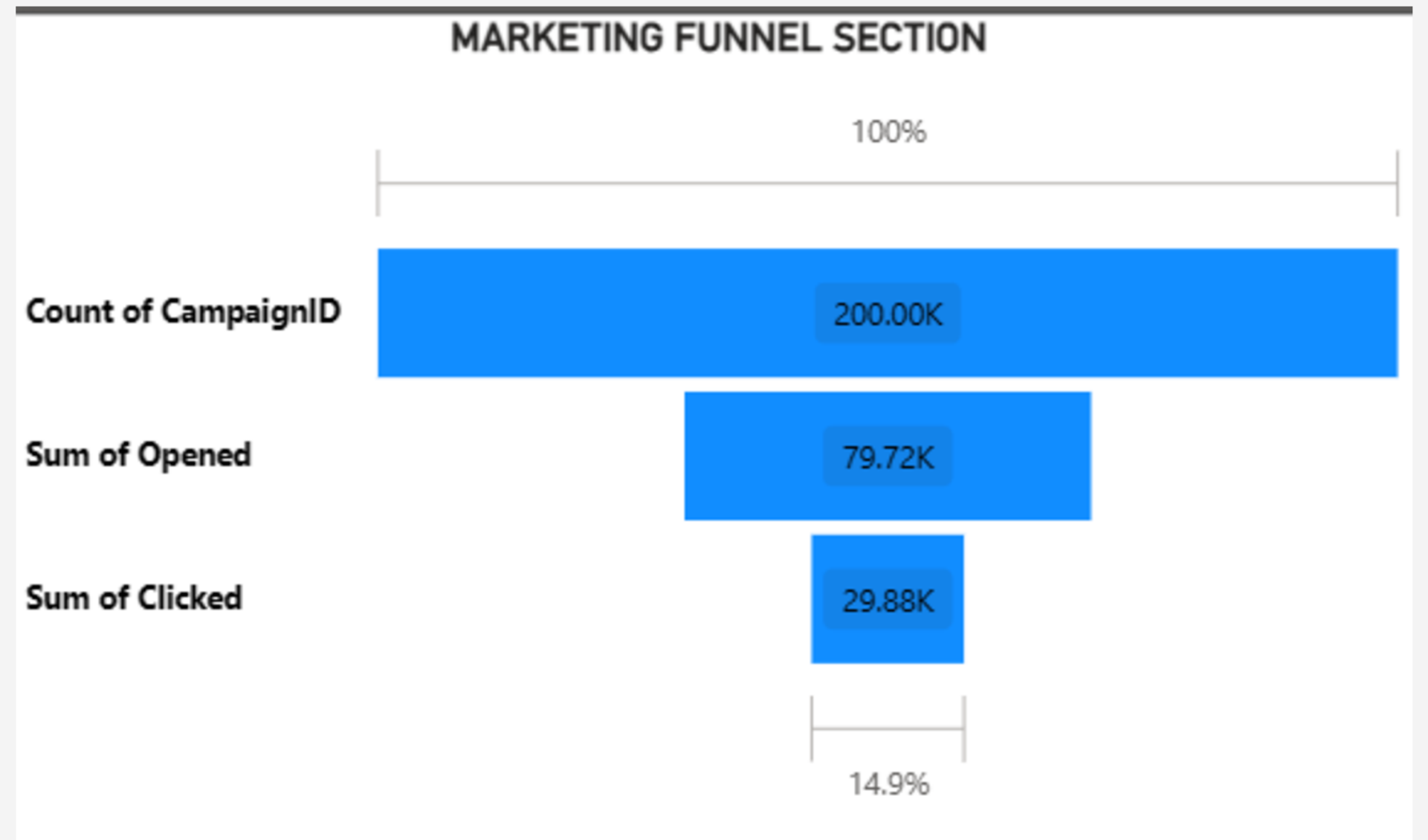
- Relevant marketing content
- Good conversion potential
- Effective campaign communication



MARKETING FUNNEL ANALYSIS

Funnel Stages

- 1.Campaign Sent
- 2.Campaign Opened
- 3.Product/Link Clicked



KEY BUSINESS INSIGHTS

Major Findings

-  Strong customer engagement
-  High campaign open rate
-  Good click-through performance
-  Customers interact repeatedly across campaigns

Business Recommendations

- Continue targeted campaigns
- Improve click conversion strategies
- Add personalized recommendations
- Track campaign timing and customer demographics in future datasets

TOOLS & TECHNOLOGIES

Tools Used

- Python
- Pandas
- NumPy
- Jupyter Notebook

Skills Demonstrated

- Data Cleaning
- EDA
- Data Visualization
- Business Analytics
- Marketing Analytics

CONCLUSION

This project successfully analyzed customer engagement behavior using Python-based EDA techniques.

The analysis revealed:

- Strong open rates
- High customer interaction
- Good campaign performance
- Valuable insights for marketing optimization

Future Improvements

Future datasets could include:

- Purchase history
- Customer demographics
- Device/location data
- Revenue metrics
- Campaign timestamps

